



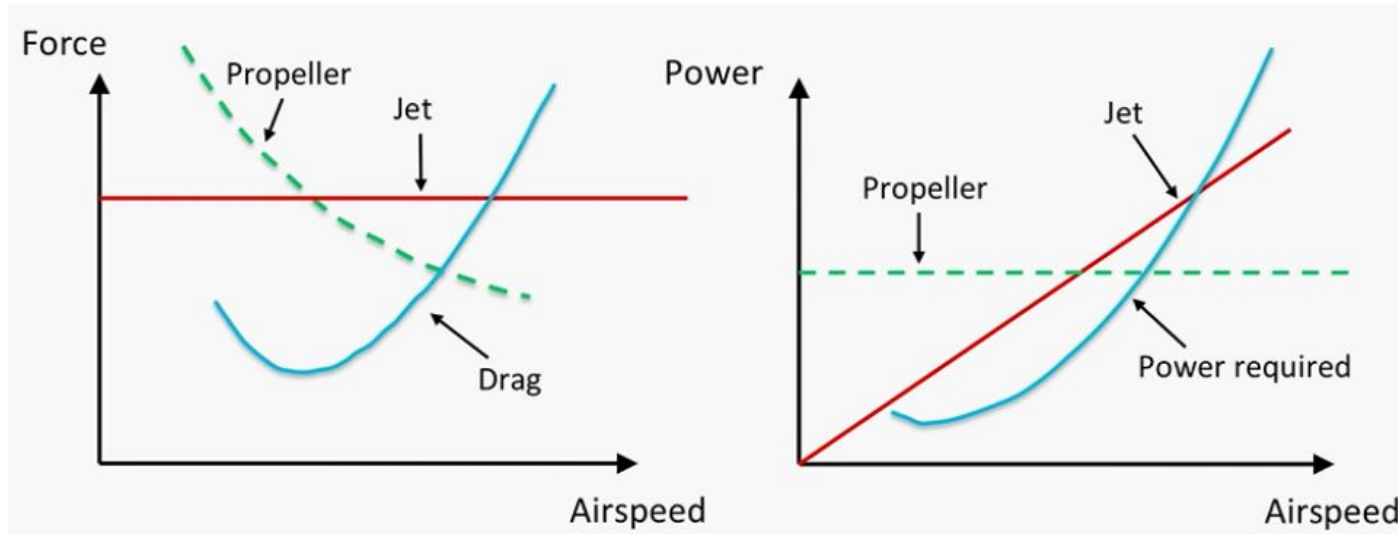
First year Aerospace Engineering

Introduction to flight mechanics

Marcos Carrera Valle

What is Flight Mechanics?

What can we do with our performance diagrams?



Minimum Speed



Maximum Airspeed



Maximum Range

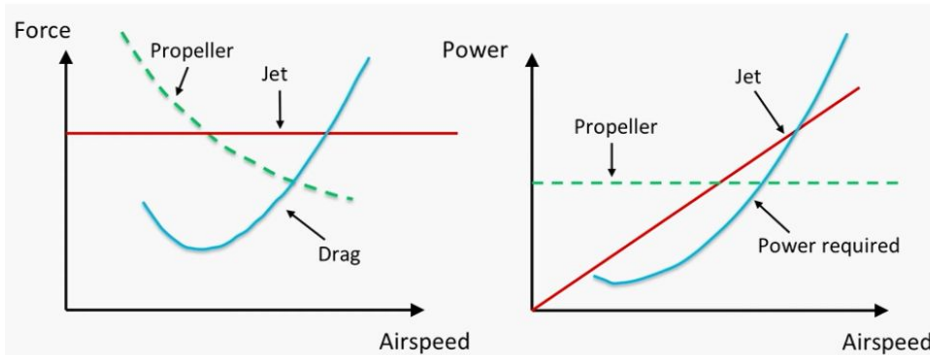
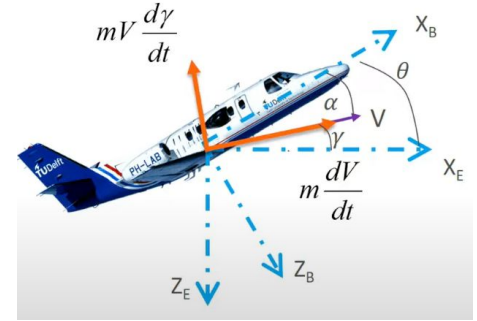


Maximum Endurance



What will we use

Both performance diagrams and equations of motion



$$\sum F_{\parallel V} : T \cos \alpha_t - D - W \sin \gamma = \frac{W}{g} \frac{dV}{dt}$$

$$\sum F_{\perp V} : L - W \cos \gamma + T \sin \alpha_t = \frac{W}{g} V \frac{d\gamma}{dt}$$

Horizontal flight - assumptions

Steady: Flight in which the forces and moments acting on the aircraft do not vary in time, neither in magnitude, nor in direction ($dV/dt = 0$)

Horizontal: The aircraft remains at a constant altitude ($\gamma = 0$)

Symmetric: flight in which both the angle of sideslip is zero and the plane of symmetry of the aircraft is perpendicular to the earth ($\beta = 0$ and the aircraft is not turning)

Simplified Equations of Motion

$$\sum F_{\parallel V} : T \cos \alpha_t - D - W \sin \gamma = \frac{W}{g} \frac{dV}{dt}$$

$$\sum F_{\perp V} : L - W \cos \gamma + T \sin \alpha_t = \frac{W}{g} V \frac{d\gamma}{dt}$$

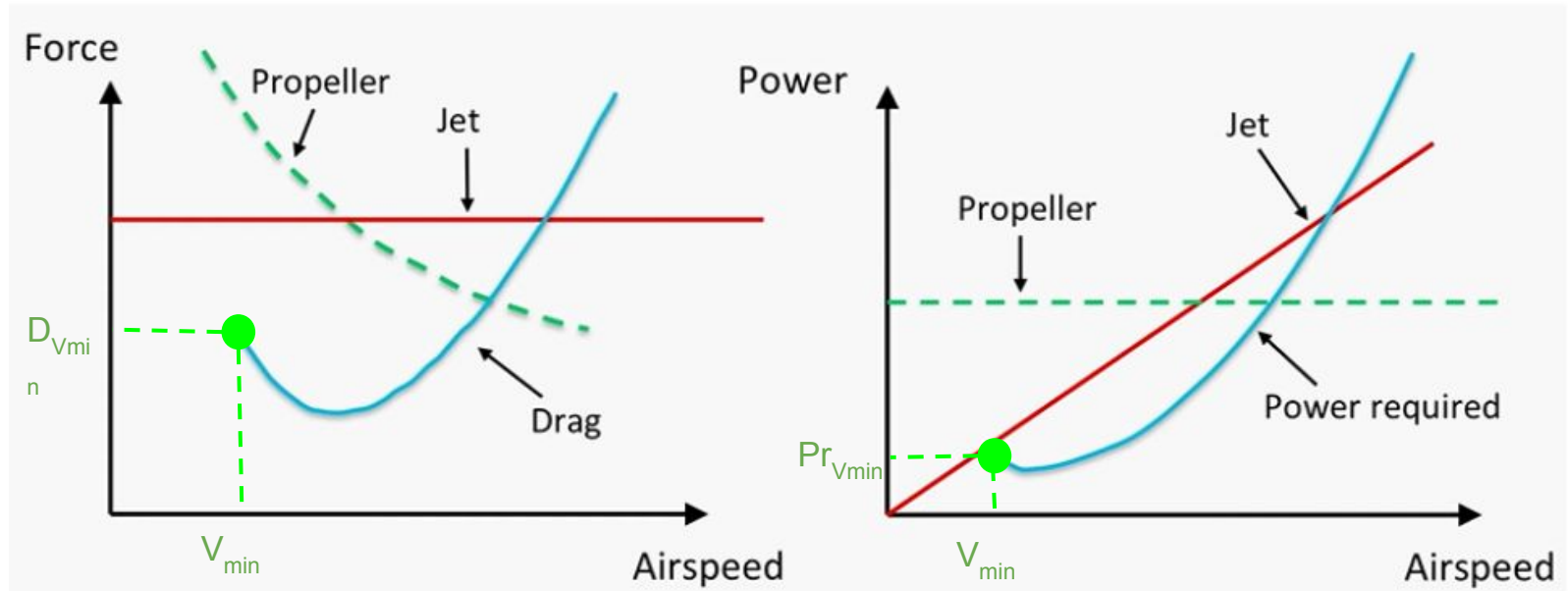
$$\sum F_{\parallel V} : T = D$$

$$\sum F_{\perp V} : L = W$$

Minimum Airspeed



Minimum airspeed



Minimum airspeed

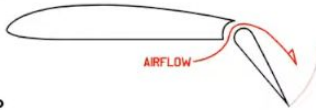
PLAIN FLAP



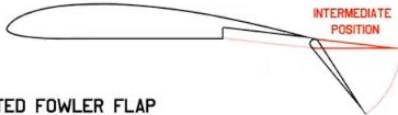
SPLIT FLAP



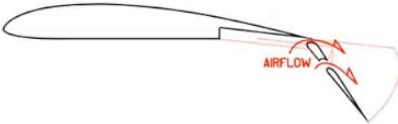
SLOTTED FLAP



FOWLER FLAP



DOUBLE-SLOTTED FOWLER FLAP



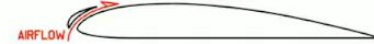
KRUEGER FLAP



LEADING EDGE DROOP



HANDLEY-PAGE SLOT





Minimum airspeed

For the following exercises we will use the Spirit of St. Louis, the first aircraft to cross the Atlantic Ocean



Spirit of St. Louis

Aircraft data:

$$C_{L_{max}} = 1.24$$

$$S = 29.68 \text{ m}^2$$

$$W_{max} = 22.8 \text{ kN}$$

$$W_{min} = 10.7 \text{ kN}$$

$$\rho = 1.225 \text{ kg/m}^3$$

Minimum airspeed

What is the minimum airspeed at minimum and maximum weight configurations?

Aircraft data:

$$C_{L_{max}} = 1.24$$

$$S = 29.68 \text{ m}^2$$

$$W_{max} = 22.8 \text{ kN}$$

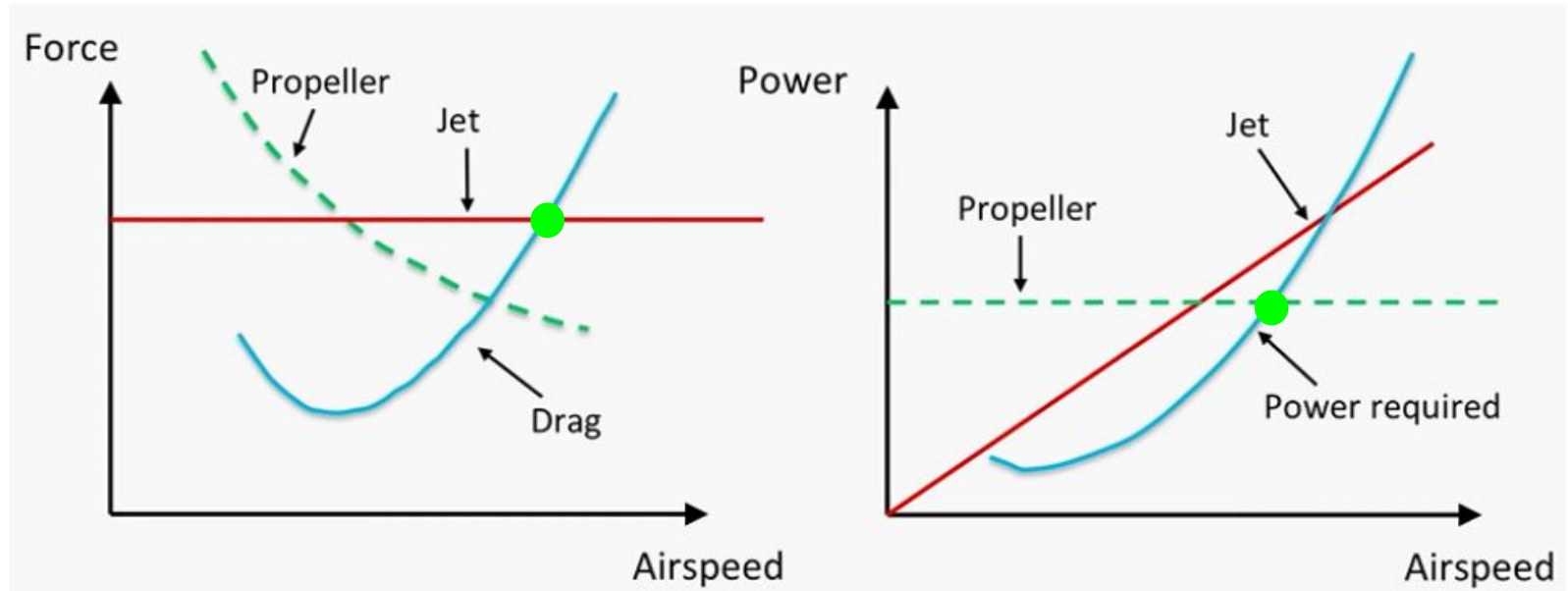
$$W_{min} = 10.7 \text{ kN}$$

$$\rho = 1.225 \text{ kg/m}^3$$

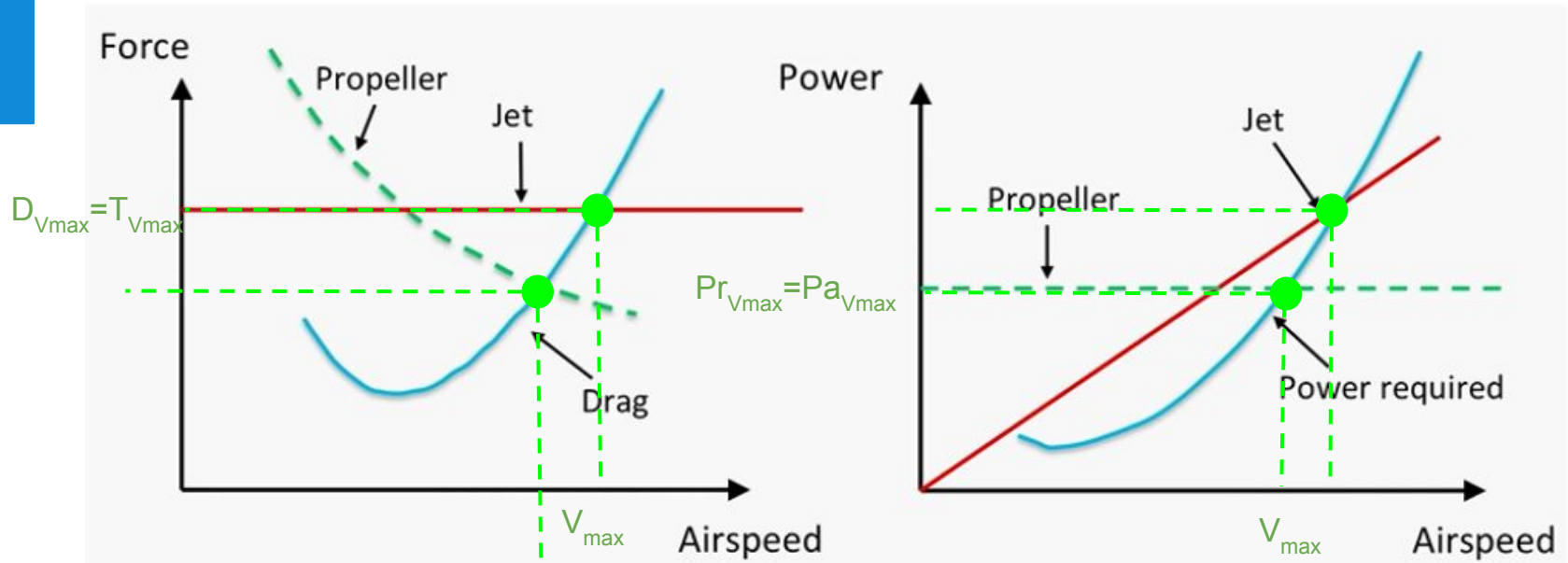
Maximum Airspeed



Maximum airspeed



Maximum airspeed



Maximum Range

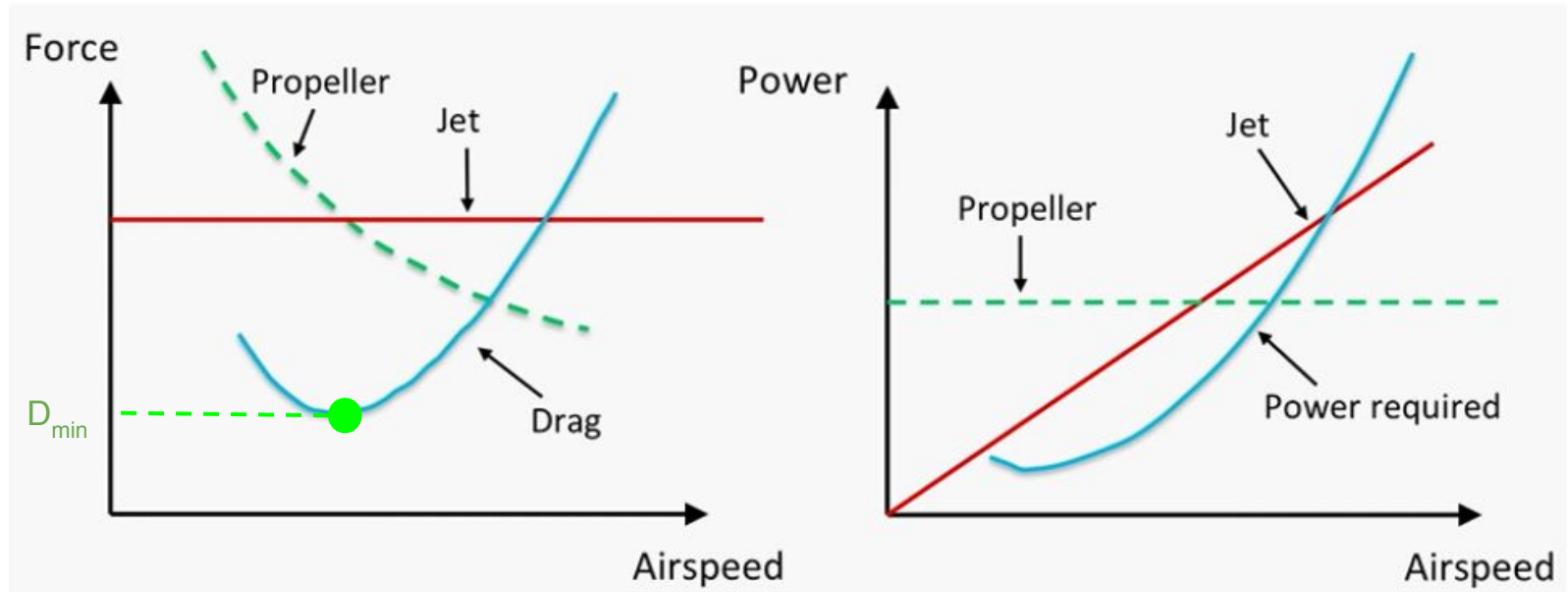


Maximum range

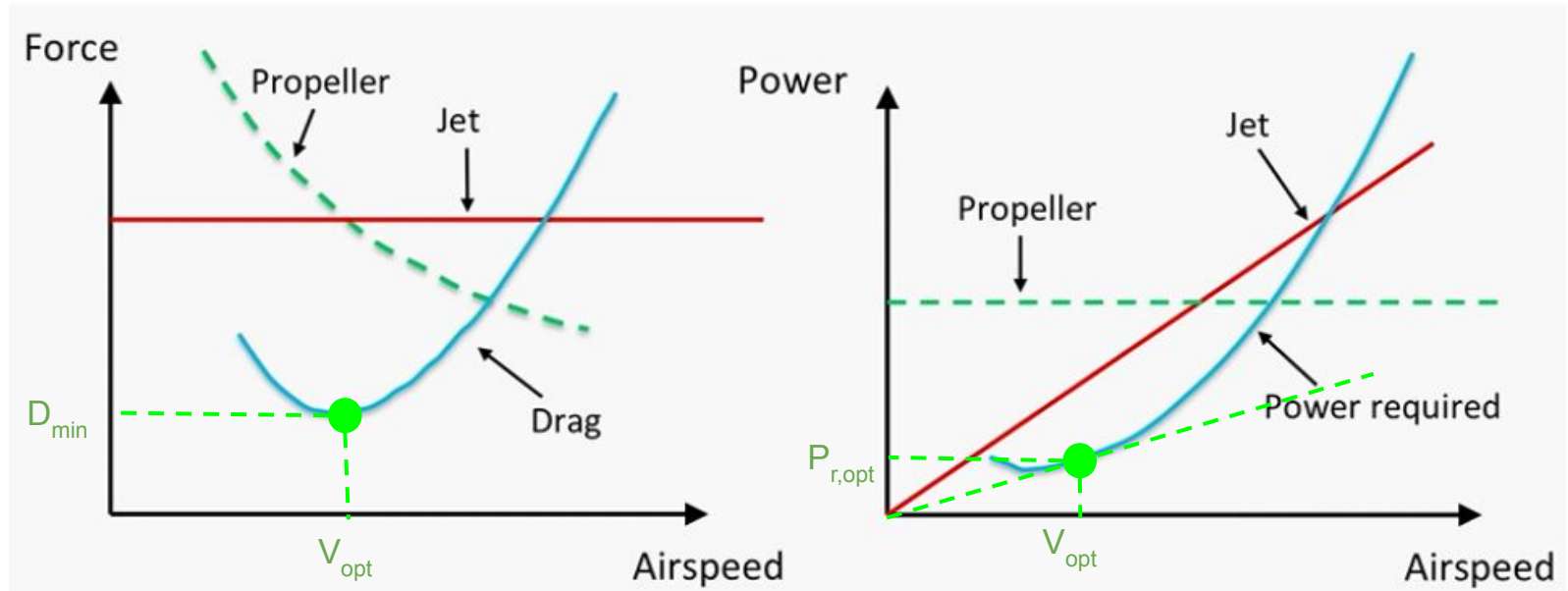
All about the fuel available!

How efficiently we use fuel

Maximum Range

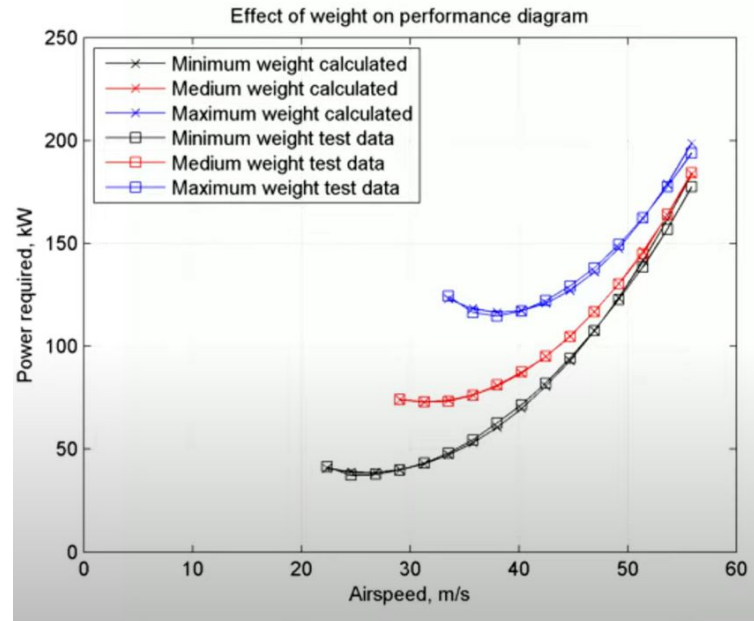


Maximum Range



Maximum Range - Effect of weight

Power required decreases as mass decreases

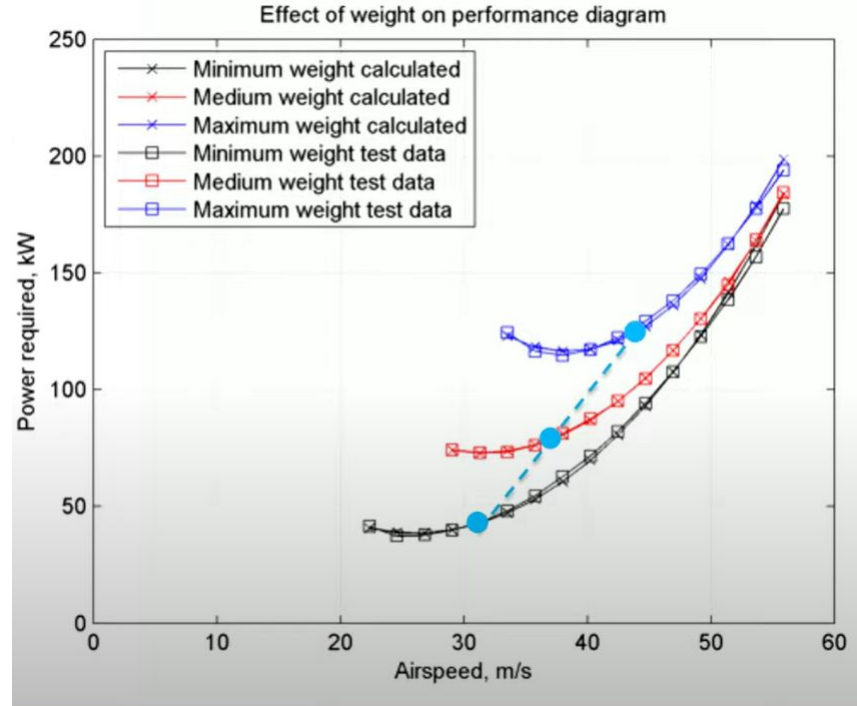


Maximum Range - Effect of weight

Optimal speed changes throughout mission

At the start, speed is quite high

As flight progresses, speed decreases



Maximum Range

Let's continue with our aircraft
Obviously range was important for this mission



Aircraft data:

$$C_{D_0} = 0.069$$

$$k_2 = 0.168$$

$$S = 29.68 \text{ m}^2$$

$$W = 10.742 \text{ kN}$$

$$\rho = 1.225 \text{ kg/m}^3$$

Maximum range

Calculate the optimal lift coefficient and the optimal velocity?

What is the total drag on the aircraft at this configuration?

What is the specific range?

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$$C_{D_0} = 0.069$$

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$$\rho = 1.225 \text{ kg/m}^3$$

$$\eta_j = 0.75$$

$$c_p = 6.45 \cdot 10^{-8} \frac{\text{kg}}{\text{Ws}}$$

Maximum range

Calculate the optimal lift coefficient and the optimal velocity?

What is the total drag on the aircraft at this configuration?

What is the specific range?

```
CLopt: 0.6408699444616557  
Vopt: 30.36495395809254 m/s  
D: 2313.099580984788 N  
R: 5026.98071122112 m/kg
```

Aircraft data:

$$C_{D_0} = 0.069$$

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Maximum Endurance



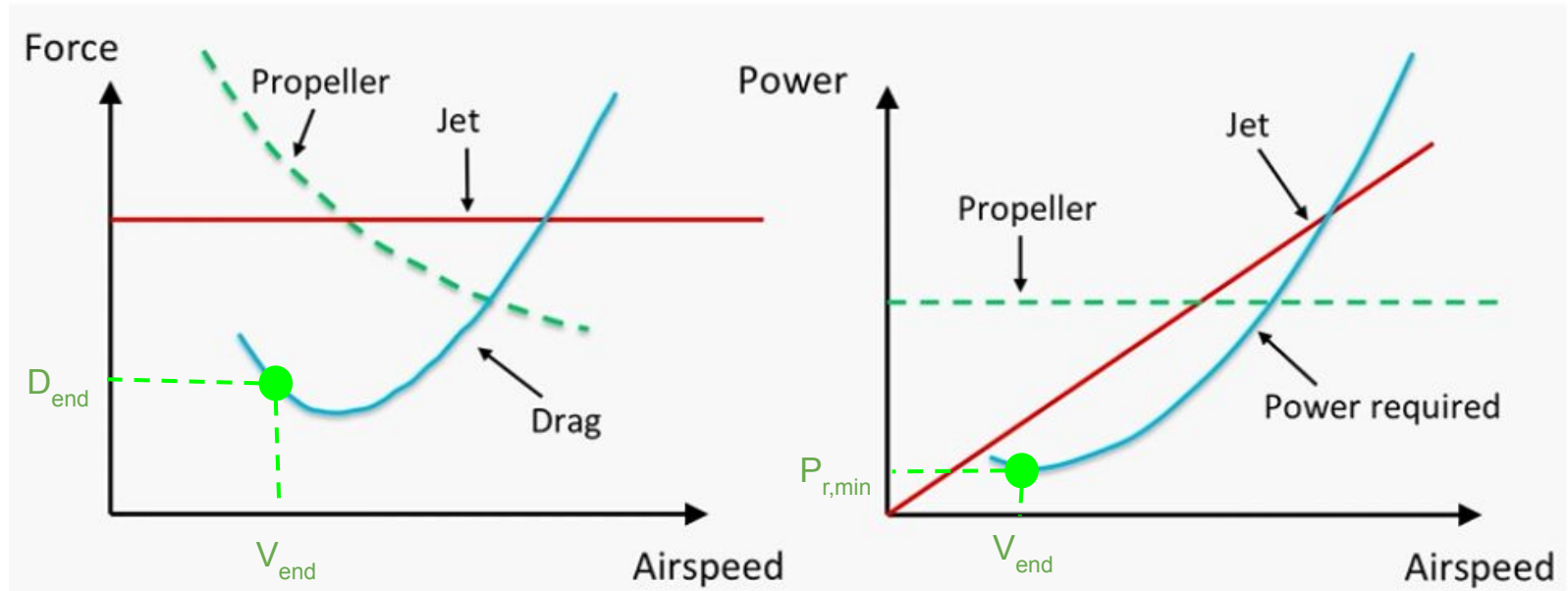
Maximum endurance

How long can you stay up in the air?

All about the fuel flow!

We want to keep fuel flow low

Maximum endurance



Homework

Let's look at one of the most sold aircraft of all time, the a320

These are the specifications:

$$W_{\text{take-off}} = 73,500 \text{ kg}$$

$$W_{\text{landing}} = 64,500 \text{ kg}$$

$$W_{\text{empty}} = 41,310 \text{ kg}$$

$$C_{L,\text{max}} = 2.2$$

$$C_{L,\text{max,HLD}} = 2.56 \text{ (With HLD / High Lift Devices)}$$

$$S = 122 \text{ m}^2$$

$$b = 34 \text{ m}$$

$$e = 0.9$$

$$C_{D,0} = 0.012$$

$$\rho_{\text{cruise}} = 0.41271 \text{ kg/m}^3$$



Homework

The aircraft approaches Schiphol Airport ($\rho = 1.225 \text{ kg/m}^3$) during landing

What is the (minimum) approach speed?

What is the power required for this manoeuvre?



Homework

We want to focus now on the maximum range.

For all weights given (take off, landing, and ferry weight (empty))

What is the optimal lift coefficient to achieve this?

What is the optimal velocity?

What is the drag under these conditions?

For take off and landing, use Schiphol airport density, for ferry weight use cruise density



Homework

For some reason NATO wants to use the a320 as radar aircraft for surveillance of Russian territory

What is the optimal lift coefficient for this mission?

What is the velocity at which it will fly?

